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<<organization>>

Change Management Process

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Document Control

This document is the responsibility of the Author. It is subject to formal change control after the initial approved release (e.g. version 1.0). Change requests must be communicated to the Author in writing. Agreed changes are recorded in the Version History of the document.

Version History

Date	Version	Author	Changes
18/04/20XX	0.1	John Doe	New Document – initial draft
01/06/20XX	1.0	John Doe	Issued to Business

Reviewers

Name	Role	Date
Jason Doe	Head of IT	01/06/20XX

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Overview

Change Requests

- This document outlines the process for formally managing change requests within <<organization>>. Change is defined as adding, modifying, or removing any authorized, planned or supported service or service component that could impact <<organization>> operations. Examples include:
    - Releasing new versions of AI software
    - Updates to internally managed infrastructure
    - Version increments of a dataset.
- Changes are categorized as follows:
    - 'Standard Change'
    - 'Change'
    - 'Emergency Change'
    - 'Ad hoc Change'.

Scope of Change Management

- The scope for change management is any change (IT, AI development, data management, cybersecurity) impacting <<organization>> operations. It covers all configuration items (physical CIs, virtual or other types of assets such as agreements or contracts) across the product lifecycle.
- This Change Management Process does not cover:
    - Changes to business strategy or organizational changes
    - Support requests (incidents, application support and order management)
    - Identity and access management.

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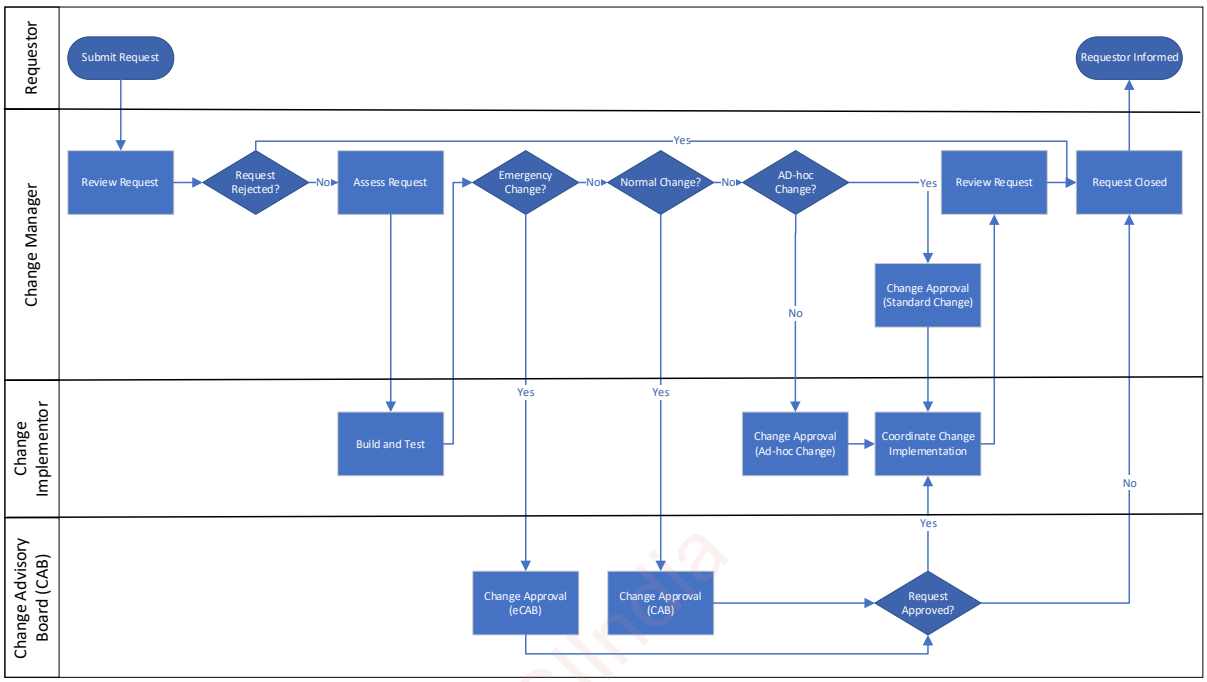
Change Management

Change Management Process

Change Management Process Flow



Change Management Key Activities

Process step	Description	Request / ITSM State
Submit Request	The requester submits a request for change (RFC) via <<organization's>> ITSM tool, <<ticketing system>>. A unique ID is allocated to the request.	New
Review Request	The RFC is reviewed concerning user validation, accuracy and required approvals. If incorrectly submitted, the RFC can be rejected and closed at this stage.	Assess
Assess Request	An impact assessment is performed as defined in the Change Impact Assessment section below. The backout plan is assessed, and a priority is assigned to the RFC. Each RFC is classified as 'Standard', 'Change' or 'Emergency' (rated as Minor/Medium/Major)	Assess
Build and Test	The change implementer schedules the change in the test environment for the testing to be performed. The test results are recorded for the next phase of approval. Approval is requested, and the change status is amended in <<ticketing system>>.	Assess

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Change Approval (Standard Change)	'Standard changes' are approved without the requirement for assessment by the Change Advisory Board (CAB).	Authorize/ Scheduled
Change Approval (Ad hoc Change)	'Ad hoc changes' are approved between the implementor and requestor.	Authorize/ Scheduled
Change Approval (eCAB/CAB)	The change is discussed by the CAB, or eCAB (see below), with the key stakeholders to discuss the impact of the change, review the implementation plan, and assess any conflict/contentions with other scheduled changes. Test results are evaluated and may be sent for re-assessment if concerns are identified. The change is approved or rejected by the change manager, with the consent of the CAB, and updates made if required. The change record is updated with the results of deliberations. The change is scheduled and set for implementation. The CAB also considers if business or customer communication is required and if the RFC would require a post implementation review (PIR) to be performed. The change status is amended in <<ticketing system>> once approved.	Authorize/ Scheduled
Coordinate Change Implementation	If the CAB agrees, business and/or customer communications are issued to the customer, the business and other stakeholders as applicable. The change implementer implements the approved change as per the implementation plan and, if necessary, invokes the back-out and recovery plan after informing the stakeholders. Incidents may be raised if the change fails and/or the agreed change window/downtime is extended. The change status is updated (implemented, backed out or failed).	Implement
Review Request	The effect of the change is verified against expectations, and appropriate action is taken if the effects differ from expected. The PIR is completed if required by the CAB. Feedback is captured from the business/customer as part of the PIR.	Review
Close Request	The RFC is closed as either 'successful', 'successful with issues' or 'unsuccessful' and stakeholders are informed.	Closed/ Cancelled

Procedure

Standard Change (Service Request)

- A 'Standard Change' is a change that has been determined to have negligible risk and minimal, well-known impact and, as such, has been preauthorized by the CAB.
- The following criteria shall be met for a change to be considered a 'Standard Change':
  - There is a defined trigger to initiate the request
  - The tasks are well-known, documented and proven
  - The change is of acceptable or no risk to the organization

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- There is no service or user impact
- Approval has been given in advance, and no additional approval is required
- A back-out or remediation plan is available
- Any required budget is pre-approved.

- These are preauthorized changes that need no or low-level sign-off. The type of changes that can be performed as a standard change is defined for each IT Team, the Data Management Team, and the AI Development Team.
- Note that the Cybersecurity Team does not perform standard changes since none of the changes undertaken are considered negligible risk.
- These lists of changes are not exhaustive. Sometimes, judgment must be used to identify the correct method of change. It should be noted that any change conducted without the appropriate management sign-off will be defined as an unauthorized change. A change that is to be categorized as a 'Standard Change' must not:
  - Affect live services
  - Make any addition or removal of services or hardware from the <<organization>> domain
  - Affect any supporting child or parent applications
  - Result in any service outage or downtime.
- A 'Standard Change' must:
  - Have the ability to be undone with zero impact
  - Be supported by a how-to guide.

<u>Process for Submitting a 'Standard Change' Request.</u>

- The 'Standard Change' requestor submits a Standard Change Template to the Change Management Team, which verifies the Standard Change Template for completeness, accuracy, and compliance with the risk, impact, and 'Standard Change' criteria.
- Once approved and added to the Standard Change Register, the change can be executed, and future changes that meet the agreed definition are completed as a 'Standard Change'. If the Standard Change Proposal is rejected, it shall be processed as a (normal) 'Change'.
- If a Standard Change requires software development activities, the change is passed to the AI Development team, with development work and testing undertaken in accordance with the AIMS0006 AI Development Process.

Change

- A 'Change' is a level above a 'Standard Change' and sometimes results in an action that could negatively impact <<organization>> operations. Managing a 'Change' is important to minimize downtime and disruption and to log changes that have been made to help diagnose other issues or track the changes in the future. All changes

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need to be approved at the weekly CAB meeting. When the change is proposed, it must be supported with the following:

- Detailed description of the change
- Information on why the change needs to take place
- A plan for the implementation of the change
- Risks associated with the change
- Time scales of when the change is to take place
- A roll back plan.

- At the CAB meeting, service managers present and discuss changes that they wish to have approved. Collective approval or rejection will be determined at the CAB meeting. The outcome of this meeting will be recorded in the Approved Change Register. Once a change has been authorized, it may occur under the approved schedule. All outstanding, unapproved changes are reported weekly to the service managers.
- Examples of a 'Change' include:
  - Releasing new versions of AI software
  - Version increments of a dataset
  - Changes to settings that affect an application beyond user customization
  - Any change to a firewall, switch or server
  - New hardware installations such as CCTV cameras, switches or firewalls.
- A change request is submitted to the CAB by creating a ticket in <<ticketing system>>. The request is initially entered as an unapproved change, and the source should correspond to one of the service areas:
  - AI Development
  - Data Management
  - Cybersecurity
  - IT Networks
  - IT Operations.
- When a change is made, it must be in line with the agreed details in the change request. Other changes are not to be considered or made without approval from the CAB.
- If a Change requires software development activities, the change is passed to the AI Development team, with development work and testing undertaken in accordance with the AIMS0006 AI Development Process.

Emergency Change

- In some cases, a change will have a degree of urgency and may need to be made without the prior approval of the weekly CAB meeting. In such circumstances, an 'Emergency Change' record shall be created in line with the (normal) change guidance to the point of implementation.

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- At the earliest opportunity after the 'Emergency Change' is made, an Emergency CAB (eCAB) meeting shall be convened to review the change retrospectively.
- Changes of this nature shall only be made during a critical incident, which impacts or could impact multiple staff or areas, and where the delay pending a normal CAB meeting is not viable.
- In the case that the eCAB determines that an emergency change made does not meet the criteria of an 'Emergency Change', the change must be backed out immediately and be submitted as a (normal) Change for consideration at a CAB meeting.
- Note all incidents, including critical incidents resulting in an Emergency Change, are recorded and managed in accordance with the <<reference>> Incident Management Process.
- The nature of an Emergency Change is such that it will require a rapid resolution, perhaps with a manual workaround, but this may require a more permanent solution involving software development. In this case, a 'normal' Change will need to be submitted for the permanent solution, which will subsequently be passed to the AI Development team to complete development work and testing in accordance with the AIMS0006 AI Development Process per above.

Ad hoc Change

- From time to time, it is necessary to make minor changes that have no impact on production operations, such as correction of a print queue name that may have been entered erroneously during setup. Such changes are termed ad hoc changes and require no approval other than by the requester and implementer. Ad hoc changes are infrequent but a <<ticketing system>> ticket must be raised, and though they do not require CAB approval ad hoc change tickets are reviewed as part of CAB meetings to monitor frequency of such changes.

Change Impact Assessment

Business Risk Assessment

- Business risk is assessed through consideration of the following criteria:
  - **Business Impact**: The negative effect of the change on the user base, sites, and services
  - **Downtime**: The duration that systems, sites or services will be unavailable
  - **Scheduling**: The negative effect of the change on agreed service level agreements (SLAs) or maintenance windows.
- The business risk assessment criteria are assessed in accordance with the Business Risk Matrix as set out in Table 1.

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Table 1: Business Risk Matrix

Value	Business Impact	Downtime	Scheduling
3	· High visibility · High number of users · Business critical services · Company-wide impact	· Downtime that has a risk of impacting business-critical client commitment(s) · Requires downtime for the test window · Work is held during testing	· Unavoidable breach of agreed SLAs · Outside established maintenance windows
2	· Minimal visibility · Minimal number of users · Minimal number of sites/services · Minimal company-wide impact	· Minimal downtime · Requires no downtime for test window · Work is not held during testing	· Agreed upon an exception to SLAs
1	· Low visibility · Low number of users · Single site/service · Change is transparent to the users and/or business	· No downtime · No user testing is needed	· No impact to agreed SLAs

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Technical Risk Assessment

- Technical risk is assessed through consideration of the following criteria:
  - **Technical Impact:** The negative impact on systems, sites and/or services if the change is not implemented.
  - **Back-out:** The availability of systems, sites and/or services and the complexity of reverting to the pre-change state.
  - **Complexity:** The level of difficulty and effort required to implement the change.
- The technical risk assessment criteria are assessed in accordance with the Technical Risk Matrix as set out in Table 2.

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Table 2: Technical Risk Matrix

Value	Technical Impact	Back-out	Complexity
3	· Business critical systems/services will be or may be unavailable, unstable or vulnerable	· Difficult or complex · More than 1 hour to back-out · Requires a system/service downtime · Will impact related systems/services	· Highly complex and detailed effort to implement
2	· Non-business systems/services may be unavailable, unstable or vulnerable	· Moderate effort · Less than 1 hour to back-out · Little impact on system/service availability during back-out	· Moderately complex and detailed effort to implement
1	· Transparent or no impact to any system/service	· No impact on system/service availability during back-out	· Easy and minimal effort to implement

Change Impact Assessment

- The business and technical impacts are combined to formulate an overall change impact assessment, as set out in Table 3.

Table 3: Change Impact Matrix

Business Risk	3	Medium	High	High
	2	Low	Medium	High
	1	Low	Low	Medium
		1	2	3
	Technical Risk			

- Note:
  - All ‘High’ and ‘Medium’ changes shall be assessed and authorized by the CAB only.
  - The CAB shall decide on changes that must follow a post-implementation review.
  - ‘Low’ changes shall be assessed and authorized outside the CAB meetings by the change manager or delegated approvers.

Change Priority

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- A change analyst assesses and updates if needed, the initial priorities set out in the Requester's risk assessment form and updates <<ticketing system>> as needed. Change priorities are:
    - **Low**: A low-priority change is typically described as a desirable, but not a necessary, change
    - **Medium**: Usually a non-time-critical change that provides the opportunity to be thoroughly planned
    - **High**: A change typically required to resolve a related incident or underlying problem/error.

Change Freezes

- At certain business critical times, such as during financial year end processing, it is necessary to avoid making changes to critical systems by implementing a change freeze. A change freeze can be requested by any department within <<organization>>, either specifically relating to that department of across the whole organization.
- A change freeze is request by submitting a change freeze proposal form for consideration by the CAB.

Change Advisory Board (CAB)

- The Change Advisory Board (CAB) comprises nominated individuals who support the assessment, prioritization, authorization, and scheduling of changes. The CAB shall also review the following when submitted:
    - Standard change proposal
    - Change freeze proposal.
- CAB membership is set out in Table 4.

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Table 4: CAB Membership

CAB Role	Position	Name
Primary attendee (CAB Chair)	Head of IT	Jason Doe
Primary attendee	Head of IA Development	Jonathan Doe
Primary attendee	Change Manager	Jaclyn Doe
Primary attendee	Data Management Lead	Jeremy Doe
Primary attendee	Quality Assurance Lead	Jade Doe
Primary attendee	Cybersecurity Lead	Jake Doe
Primary attendee	AI Ethics Lead	Jim Doe
Optional attendee	Chief AI Officer	Jacinda Doe
Optional attendee	AI Strategy Head	Justin Doe
Optional attendee	Risk and Compliance Lead	Jane Doe
Optional attendee	Innovation Lab Director	Jo Doe
Optional attendee	Herad of Human Resources	Jamal Doe
Optional attendee	CFO	Jarred Doe
Optional attendee	Head of Procurement	Jenny Doe
Optional attendee	Sales and Marketing	Jas Doe
Optional attendee	Service Desk Manager	Jean-Paul Doe

- For the CAB to convene and approve a submitted change, at least one primary attendee shall be present. If no primary attendees can attend, then the CAB must be rescheduled.
- Membership of the eCAB is the same as a regular CAB, but attendance is on a best endeavours basis.

Change Advisory Board Schedule

- The CAB shall convene every Wednesday at 10:00 am and review requests submitted for approval before 17:00 on the preceding Monday, as follows:
  - 'Standard Changes' are not discussed
  - All (normal) 'Changes' shall be discussed
  - All service request model submissions shall be discussed.
- Changes submitted after 17:00 on the preceding Monday cannot be approved or implemented before discussion by the CAB on the following Wednesday, unless classified as an 'Emergency Change'.
- The eCAB convenes on an as-needed basis for consideration of 'Emergency Changes'.

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Change Advisory Board Meeting Template

1. Meeting details
    1.1. When:
    1.2. Where:
    1.3. Frequency:
    1.4. Format:
2. Attendees:
    2.1. Primary members

    Name, title, role, email address, contact number - present/ absent/ delegated

    2.2. Optional members

    Name, title, role, email address, contact number - present/ absent/ delegated
3. Review of previous meeting action items
    3.1. Action item, assigned to, planned action date, status, update, remarks
4. Meeting details
    4.1. Review of changes planned in the next seven days
    4.2. Review of failed changes in past seven days
    4.3. Review of ad hoc changes implemented in past seven days
    4.3. Review of 'Standard Change' proposals
    4.4. Review of (normal) 'Change' proposals
    4.5. Review of change freeze proposals
5. Meeting action items
    5.1. Action Item, assigned to, planned action date, status, update, pemarks
    5.2. Forward schedule of change (FSC)

Business and Customer Communication

Overview

- The business and customer communication requirements shall be defined during the CAB approval process for a submitted request.
- If required, business communications shall conform to the defined template (per below in the Business Communication Template). The Service Desk shall issue communications in advance of the scheduled change implementation.

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- If required, customers are informed of changes following contractual / license terms. Communications are handled by the affected customers' sales account manager.

Business Communication Template

Sending Mailbox: IT Services Communications (IT.comms@<<organization>>.co.uk)

Recipient Distribution List: All <<organization>> Group Users (all<<organization>>staff@<<organization>>.co.uk)

Subject: IT Change Notification: *Impacted Service* *Notification Type*

IT Change Notification: ****Notification Type****	
Impacted Service	
Change Details:	
Change Start Time:	Monday 1 January 20XX 00:00 GMT
Change End Time:	Monday 1 January 20XX 01:00 GMT
Business action Required:	

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